# Inorganic Cumulative Examination
## April 6, 2012

The ability to synthesize small molecules can be important in mechanistic studies, isotopic labeling and the synthesis of larger molecules. In addition, it provides a fundamental look at some of the chemistry. In that context, provide a small-scale laboratory synthesis using common laboratory reagents for the following small molecules. Getting the material from a lecture cylinder is not considered synthesis. Direct combination of the elements is rarely a good laboratory route to a compound, e.g., making ammonia from hydrogen and nitrogen is not a viable option.

1. $H_2$
2. $O_2$
3. $N_2$
4. $Cl_2$
5. $NH_3$
6. CO
7. $N_2O$
8. NO
9. $CO_2$
10. HCl
11. $NO_2$
12. $SO_2$
13. $PH_3$
14. $H_2S$
15. $B_2H_6$